

Philosophy 106-14: Introduction to Logic

Cal State University, Fullerton

CPAC 148 - Lecture Room

MoWe 8:30AM - 9:45AM

Instructor: Matthew Madrugá, email: mmadruga@fullerton.edu

Office Hours: 11:30-1:30 Monday, H-212A or by appointment

Zoom: <https://fullerton.zoom.us/j/5635533673>

Required Texts(Free/Open Access):

1.) *Introduction to Logic* by Ryan Simonelli

https://www.ryansimonelli.com/uploads/1/3/3/4/133499356/simonelli_-_introduction_to_logic_05.pdf

2.) *An Introduction to Formal Logic* by P.D. Magnus

<https://www.fecundity.com/codex/forallx.pdf>

3.) *An Introduction to Formal Logic* by Peter Smith

https://www.logicmatters.net/resources/pdfs/IFL2_LM.pdf

Course Description:

Logic is often mistaken for a narrow or abstract discipline, but in truth it is the foundation of responsible reasoning in every domain of life. This course introduces students to the principles of valid inference, the tools of formal proof, and the methods for evaluating arguments with clarity and rigor. Alongside these technical skills, we will examine the broader significance of logic: its role in protecting us from error, in grounding fair and consistent judgment, and in fostering accountability in our reasoning. From ethical debates to legal decisions to everyday communication, the ability to recognize good and bad arguments has profound epistemic and moral implications. Logic is not just about symbols and rules; it is about cultivating intellectual virtues: precision, fairness, and honesty; that matter whenever we claim to know, to decide, or to act.

Course Learning Outcomes:

By the end of the semester, students should be able to:

- Identify arguments in natural language, distinguish premises from conclusions, and assess their strength.
- Explain the distinctions between deductive and inductive reasoning and identify common logical fallacies.
- Recognize and articulate common barriers to clear and rational thought, including bias, ambiguity, and emotive influence.
- Demonstrate basic competence in constructing, articulating, and defending one's own views while remaining charitable toward opposing positions.
- Apply logical tools to real-world contexts, showing how rigorous reasoning has ethical and epistemic implications beyond the classroom.

GE Learning Outcomes:

- This course meets the requirements for General Education Category 1B(2025 and after), and A3(2021 and after)Students taking courses in subarea 1B(2025 and after) A3(2021 and after) will:
Understand the role of logic and its relation to language.
- Understand elementary inductive and deductive processes, including formal and informal fallacies.
- Distinguish propositions and statements of fact from issues of judgment or opinion.
- Develop skills to advocate for ideas in a reasoned, evidence-based manner.
- Reach well-supported factual and judgmental conclusions and successfully advocate for them.
- Evaluate, critique, and analyze the quality and sufficiency of evidence and reasoning in support of positions, including recognition of underlying argumentative structure.

Student Learning Outcomes:

Philosophy is distinctive in cultivating both critical and creative dimensions of thought. By the end of this course, students will:

- Improve their ability to assess the reasons for and against a position, sharpening skills in identifying valid and fallacious reasoning.
- Strengthen their capacity to consider new positions charitably and to engage in problem-solving with intellectual openness.
- Gain transferable skills in critical analysis applicable to philosophical texts, as well as to coursework in other disciplines.
- Develop foundational tools for analyzing and evaluating concepts in fields such as mathematics, computer science, law, and the social sciences.
- Leave the course better prepared to contribute thoughtfully to civic, ethical, and professional discussions requiring clarity, fairness, and rigor in reasoning.

Grading:

4 Quizzes: 20%

Midterm: 30%

Final: 40%

Attendance/Participation: 10%

GRADING SCALE:

A=100-92

A-=92-90

B+=90-88
B=88-82
B-=82-80
C+=80-78
C=78-72
C-=72-70
D+=70-68
D=68-62
D-=62-60
F=<60

Weekly Schedule:

Week 1 (Aug 25, 27) – What is Logic? Arguments & Structure

- Mon (8/25): Introduction to logic; arguments, premises & conclusions
 - Simonelli Ch. 1 (What is Logic?)
 - Magnus Ch. 1 (§1.1–1.2: Arguments, Validity, Soundness)
 - Smith Ch. 1 (Logic & Argument)
- Wed (8/27): Identifying arguments in natural language
 - Simonelli Ch. 2
 - Magnus §1.3–1.4 (Identifying Arguments, Sentences vs Propositions)
 - Smith Ch. 2 (Validity, Consequence, Proof)

Week 2 (Sep 1, 3) – Deduction, Induction & Language

- Mon (9/1): No Class – Labor Day
- Wed (9/3): Deductive vs Inductive reasoning, fallacies of language
 - Simonelli Ch. 3
 - Magnus §1.5–1.6 (Deductive vs Inductive, Fallacies)
 - Smith Ch. 3

Week 3 (Sep 8, 10) – Sentential Logic Basics

- Mon (9/8): Syntax of sentential logic
 - Simonelli Ch. 4 (Statements, Connectives)
 - Magnus Ch. 2.1–2.2 (Sentence Logic Syntax)
 - Smith Ch. 4 (The Language of Propositional Logic)
- Wed (9/10): Truth tables & logical relations
 - Simonelli Ch. 4 cont.
 - Magnus Ch. 2.3–2.4 (Truth Tables, Logical Connectives)
 - Smith Ch. 5 (Truth-Functional Semantics)

Week 4 (Sep 15, 17) – Validity & Formal Proofs (Propositional)

- Mon (9/15): Testing validity with truth tables
 - Simonelli Ch. 5
 - Magnus Ch. 2.5–2.6 (Validity via Truth Tables)
 - Smith Ch. 6 (Semantic Consequence)
- Wed (9/17): Introduction to natural deduction proofs

- Simonelli Ch. 5 cont.
- Magnus Ch. 3.1–3.2 (Rules of Inference, Conditional Proof)
- Smith Ch. 7 (Proof Systems)
- **Quiz 1 (Wed)**

Week 5 (Sep 22, 24) – Proof Strategies in Propositional Logic

- Mon (9/22): Direct proofs, conditional derivations
 - Simonelli Ch. 6
 - Magnus Ch. 3.3–3.4
 - Smith Ch. 8 (Proof Techniques)
- Wed (9/24): Indirect proofs, reductio
 - Simonelli Ch. 6 cont.
 - Magnus Ch. 3.5–3.6
 - Smith Ch. 8 cont.

Week 6 (Sep 29, Oct 1) – Predicate Logic: Syntax

- Mon (9/29): Terms, predicates, quantifiers
 - Simonelli Ch. 7
 - Magnus Ch. 4.1–4.2 (Quantifiers, Variables)
 - Smith Ch. 9 (First-Order Syntax)
- Wed (10/1): Translating English to predicate logic
 - Simonelli Ch. 7 cont.
 - Magnus Ch. 4.3–4.4
 - Smith Ch. 10 (Translation Exercises)

Week 7 (Oct 6, 8) – Predicate Logic: Semantics

- Mon (10/6): Truth in predicate logic
 - Simonelli Ch. 8
 - Magnus Ch. 5.1–5.2
 - Smith Ch. 11 (Semantics of Quantifiers)
- Wed (10/8): Validity and models
 - Simonelli Ch. 8 cont.
 - Magnus Ch. 5.3–5.4
 - Smith Ch. 11 cont.

• **Quiz 2 (Wed)**

Week 8 (Oct 13, 15) – Proofs in Predicate Logic

- Mon (10/13): Rules for quantifiers
 - Simonelli Ch. 9
 - Magnus Ch. 6.1–6.3
 - Smith Ch. 12 (First-Order Proofs)
- Wed (10/15): Strategies with quantifiers
 - Simonelli Ch. 9 cont.
 - Magnus Ch. 6.4–6.5
 - Smith Ch. 12 cont.

Week 9 (Oct 20, 22) – Midterm & Review

- Mon (10/20): Review of propositional & predicate logic
- **Wed (10/22): Midterm Exam**

Week 10 (Oct 27, 29) – Identity & Relations

- Mon (10/27): Logic with identity
 - Simonelli Ch. 10
 - Magnus Ch. 6.6–6.7
 - Smith Ch. 13 (Identity)
- Wed (10/29): Relations & multiple quantifiers
 - Simonelli Ch. 10 cont.
 - Magnus Ch. 6.8–6.9
 - Smith Ch. 13 cont.

Week 11 (Nov 3, 5) – Informal Fallacies & Argument Analysis

- Mon (11/3): Common fallacies, barriers to reasoning
 - Simonelli Ch. 11
 - Magnus Ch. 7.1 (Fallacies, Informal Reasoning)
 - Smith Ch. 14 (Limits of Formalization)
- Wed (11/5): Extended practice in identifying fallacies
 - Simonelli Ch. 11 cont.
 - Magnus Ch. 7.2–7.3
 - Smith Ch. 14 cont.
- **Quiz 3 (Wed)**

Week 12 (Nov 10, 12) – Metalogic: Soundness

- Mon (11/10): Soundness theorem overview
 - Simonelli Ch. 12
 - Magnus Ch. 7.4
 - Smith Ch. 15 (Soundness)
- Wed (11/12): Applications of soundness
 - Simonelli Ch. 12 cont.
 - Magnus Ch. 7.5
 - Smith Ch. 15 cont.

Week 13 (Nov 17, 19) – Metalogic: Completeness & Limits

- Mon (11/17): Completeness theorem
 - Simonelli Ch. 13
 - Magnus Appendix A
 - Smith Ch. 16 (Completeness)
- Wed (11/19): Limits of logic, Gödel's incompleteness
 - Simonelli Ch. 13 cont.
 - Magnus Appendix B
 - Smith Ch. 17 (Incompleteness)

Week 14 (Nov 24, 26) – Applied Logic

- Mon (11/24): Logic in philosophy, science, ethics
 - Simonelli Ch. 14
 - Magnus Ch. 8 (Applications)
 - Smith Ch. 18 (Why Logic Matters)
- Wed (11/26): Practice with applied arguments
 - Review selections from all texts
- **Quiz 4 (Mon)**

Week 15 (Dec 1, 3) – Review & Integration

- Mon (12/1): Cross-text review: key concepts and skills
- Wed (12/3): Capstone: logic's epistemic & ethical significance

Week 16

- Mon Dec 8: Final exam review; open Q&A.
- Wed Dec 10: Last class — reflection on logic's epistemic & ethical importance.

Final: Wednesday, December 17th, 9:00-10:50am

CAMPUS RESOURCES:

University Learning Center The goal of the ULC is to provide all Cal State Fullerton students with academic support in an inviting and contemporary environment. The staff of the University Learning Center tutors students, offers workshops, provides collaborative learning opportunities and assists students with their academic assignments, general study skills, and computer user needs. Location: MH 78 and MH 33. Phone: (714) 278-4349. URL: <http://www.fullerton.edu/ulc/>

The Writing Center:

The Writing Center offers all Cal State Fullerton students the opportunity to receive writing assistance from student tutors. All tutors have been selected after an application and review process. They must then complete a semesterlong course that trains them to use a variety of tutoring strategies and teaches them about responding to and evaluating writing. In half hour long tutorials, the students who come to the Writing Center will work with a tutor to create and/or improve specific assignments and, more importantly, to improve overall writing skills. Location: MH 45, in the basement of McCarthy Hall. Phone: (714) 278-3650. URL: <http://hss.fullerton.edu/english/wc/>

COUNSELING AND PSYCHOLOGICAL SERVICES:

CAPS offers free and confidential briefcounseling, assessment and referral, and crisis intervention designed to assist Cal State Fullerton students with personal or psychological problems that interfere with their educational progress. Here students can seek counseling for depression, anxiety, stress symptoms, as well as relationship, family or situational problems. The counselors at CAPS are Licensed Psychologists. Location: SHCC East Building, next to the Health Center. Phone: (714) 278-3040. URL: <http://www.fullerton.edu/shcc/CAPS%20AboutUs.htm>

Emergency Action Please see the following link regarding emergency action in the face of earthquakes, hurricanes, dust storms and other events. ALERT Emergency information URL: http://www.fullerton.edu/emergencypreparedness/ep_students.html
Disabled Student Services The mission of the Office of Disabled Student Services at California State University, Fullerton is to increase access and retention for students with permanent and temporary disabilities by ensuring equitable treatment in all aspects of campus life. The program acts as a catalyst and agent for compliance with Federal and State laws and regulations mandating equal opportunity and access for persons with disabilities. Disabled Student Services provides co-curricular and academically related services that empower students with disabilities to achieve academic and personal self-determination. Location: University Hall 101. Phone: 714-278-3117. TDD: 714-278-2786. URL: <http://www.fullerton.edu/DSS/>

Academic Integrity Statement:

Students are expected to uphold the highest standards of academic integrity. University policies on academic dishonesty will be fully enforced; please review the official policy here: [UPS 300.021](#) Academic dishonesty includes “obtaining or attempting to obtain credit for work by dishonest, deceptive, fraudulent, or unauthorized means.” The most common violation is plagiarism, including copying material from the internet. Even a single borrowed sentence, if not placed in quotation marks with proper citation, constitutes plagiarism. Any student who engages in academic dishonesty will receive a failing grade in the course, and the matter will be referred to the Dean of Students Office. If you are uncertain about how to cite sources, consult the instructor.

Class Attendance and Participation:

Regular attendance and active participation are essential for success in this course. Students may miss up to three classes without explanation. Beyond this, each unexcused absence will lower the final grade by half a letter. To excuse additional absences, appropriate documentation is required (note: work conflicts do not qualify). Participation includes contributing to discussions, asking and answering questions, and giving your full attention in class. Together, students and the instructor share responsibility for making the course engaging, thoughtful, and intellectually rewarding.

Technical Problems:

If you encounter any technical difficulties, contact the instructor immediately to document the problem. Then, contact: [student IT help desk](#), [email](#), phone = 657-278-8888, walk-in [student genius center](#), online chat - log into [portal](#); click “Online IT Help”; click “Live Chat.”

For issues with Canvas: Canvas Support = 657-278-8888, [search the CSUF Canvas Guides with AI Assistant](#), [report a technical problem](#).

University’s Disabilities Policy:

“The University requires students with disabilities to register with the Office of Disability Support Services (DSS), located in UH-101 and at (657) 278-3112, in order to receive prescribed accommodations appropriate to their disability. Students requesting accommodations should inform the instructor during the first week of classes about any disability or special needs that may

require specific arrangements/accommodations related to attending class sessions, completing course assignments, writing papers or quizzes/tests/examinations.”

Special Needs Statement:

If you have a disability or special need for which you are or may be requesting an accommodation, please inform me and contact the Office of Disability Support Services, located in University Hall 101, as early as possible in the term. For more information, the Office of Disability Support Services can be reached by calling (657) 278-3117 or visit their website at www.fullerton.edu/DSS/. Confidentiality will be protected.

Sexual Harassment Policy:

This course, its instructor and teaching assistants will comply with the University Policy Statement concerning sexual harassment to be found in UPS 240.100 (<http://www.fullerton.edu/senate/documents/PDF/200/UPS240-100.pdf>).